

Sparse Equation Discovery for a Reduced Tokamak Turbulence Model: Exact Recovery, Noise Robustness, and the Chaos Evaluation Problem

Lucas Perrier

Abstract

We apply sparse equation discovery (SINDy) to the Chen–Lin–White (CLW) reduced turbulence model, a four-dimensional chaotic system describing the interaction of a resistive tearing mode with an ion temperature gradient-driven mode in tokamak plasmas. Using a physics-informed candidate library, we show that the Sequential Thresholded Least Squares (STLSQ) algorithm achieves exact symbolic recovery—correct support and coefficient relative L_2 error below 10^{-15} —when provided clean oracle derivatives. We then systematically characterize how identification degrades across three derivative estimation regimes (oracle, numerical finite differences, and Savitzky–Golay smoothed finite differences) under six noise levels spanning four decades. Library ablation experiments demonstrate that an overly rich degree-2 library maintains exact recovery in the noiseless case but amplifies false-positive inclusions under noise, while an incomplete library missing the rational term $(PZ/S) \sin C$ floors the coefficient error at ~ 0.48 regardless of noise level. Because the CLW system is chaotic with a maximal Lyapunov exponent $\lambda_{\max} \approx 0.3$, we argue that equation recovery metrics—support identification and coefficient error—are the appropriate validation criteria, rather than long-horizon trajectory fidelity. All results are averaged over five random seeds with reported standard deviations.

Keywords: sparse identification of nonlinear dynamics, tokamak turbulence, equation discovery, chaotic systems, SINDy, Chen–Lin–White model

1 Introduction

Understanding the dynamics of turbulence in magnetically confined plasmas is a central challenge in fusion energy research [5, 9]. High-fidelity gyrokinetic simulations can capture the complex nonlinear interactions driving turbulent transport, but they are computationally prohibitive for parametric studies and real-time control. Reduced models that retain the essential physics in a low-dimensional ordinary differential equation (ODE) system offer an attractive alternative, provided the governing equations can be identified from data.

Sparse equation discovery has emerged as a principled approach to extracting interpretable governing equations from time-series data. The Sparse Identification of Nonlinear Dynamics (SINDy) framework [2] formulates the problem as sparse regression: given state trajectories $\mathbf{X}$ and their time derivatives $\dot{\mathbf{x}}$, one seeks a sparse coefficient matrix Ξ such that $\dot{\mathbf{x}} \approx \Theta(\mathbf{X})\Xi$, where $\Theta(\mathbf{X})$ is a library of candidate nonlinear functions evaluated on the data. The appeal of SINDy lies in its simplicity, interpretability, and the *a priori* imposition of parsimony through sparsity-promoting regression.

However, applying SINDy to physically relevant systems raises several practical challenges. First, derivative estimation from noisy measurements can introduce substantial errors that corrupt the regression [17, 15]. Second, the choice of candidate library is critical: too narrow a library may exclude the true dynamics, while too broad a library increases the risk of false-positive term identification [12]. Third, chaotic systems present a fundamental evaluation difficulty—even a perfect model produces trajectories that diverge exponentially from the truth at the Lyapunov timescale, making long-horizon error an unreliable metric [18].

In this work, we use the Chen–Lin–White (CLW) reduced turbulence model [5] as a controlled testbed to investigate these challenges systematically. The CLW system describes the coupling of a resistive tearing mode with an ion temperature gradient (ITG)–driven mode through four ODEs governing the tearing mode amplitude P , the sound-wave amplitude S , the ITG mode amplitude Z , and a relative phase C . Despite its low dimensionality, the system exhibits rich chaotic dynamics that are representative of the nonlinear mode coupling observed in tokamak turbulence.

We make the following contributions:

1. We demonstrate exact symbolic recovery (correct support, coefficient error $< 10^{-15}$) of the CLW equations from a physics-informed library with oracle derivatives.
2. We characterize the degradation of coefficient recovery across three derivative estimation regimes—oracle, numerical finite differences, and Savitzky–Golay smoothed finite differences (SFD)—under noise levels from $\eta = 10^{-4}$ to $\eta = 10$.
3. We perform library ablation experiments testing an extended degree-2 library and an incomplete library to quantify the sensitivity of recovery to library specification.
4. We provide a sample-size ablation and SFD hyperparameter sensitivity analysis.
5. We formalize the argument that for chaotic systems, equation recovery (support identification and coefficient error) should take precedence over trajectory-level forecasting metrics.

2 Related Work

Sparse equation discovery. The SINDy framework was introduced by Brunton, Proctor, and Kutz [2], building on compressed sensing and sparse regression. Extensions address noisy data through integral formulations [17], weak formulations [13], and ensemble methods [8]. Rudy et al. [15] extended the approach to partial differential equations. Champion et al. [3] combined SINDy with autoencoders for coordinate discovery in high-dimensional systems.

Derivative estimation. Accurate derivative estimation is a known bottleneck for equation discovery from noisy data. Total variation regularization [4], Savitzky–Golay filtering [16], and spectral differentiation [17] have been proposed. The PySINDy package [7, 11] provides a unified interface to these methods through its `SmoothedFiniteDifference` differentiator.

Reduced plasma models. The CLW system was derived by Chen, Lin, and White [5] as a reduced description of resistive tearing mode–ITG coupling in tokamak geometry. It has been used as a benchmark for nonlinear dynamics studies in plasma physics [9]. While sparse identification has been applied to other plasma-relevant systems [6, 10], a systematic study of SINDy on the CLW model with controlled noise, library, and derivative estimation variations has not been reported.

Evaluation of discovered equations in chaotic systems. The exponential divergence of nearby trajectories in chaotic systems implies that long-horizon forecast error is a poor metric for model quality [18]. Short-horizon comparisons and statistical measures (e.g., Lyapunov exponents, attractor statistics) are more appropriate [14, 19]. We build on this insight by normalizing trajectory comparisons by the Lyapunov time and focusing the evaluation on coefficient-level metrics.

3 The Chen–Lin–White System

The CLW system [5] is a four-dimensional autonomous ODE:

$$\dot{P} = P - 2 Z S \cos C, \quad (1)$$

$$\dot{S} = -\Gamma_d S + Z P \cos C, \quad (2)$$

$$\dot{Z} = -\gamma_z Z + 2 P S \cos C, \quad (3)$$

$$\dot{C} = d - \frac{P Z}{S} \sin C, \quad (4)$$

where the state variables are:

- P : tearing mode amplitude,
- S : sound-wave amplitude,
- Z : ITG mode amplitude,
- C : relative phase between the modes.

The parameters are the sound-wave damping rate Γ_d , the ITG mode growth rate γ_z , and the frequency mismatch d . Throughout this work, we fix $\Gamma_d = 2.0$, $\gamma_z = 0.8$, and $d = 2.0$, which places the system in a chaotic regime.

The system has $k^* = 8$ nonzero coefficients across the four equations: one linear term per equation for $\dot{P}$, $\dot{S}$, and $\dot{Z}$; the three cross-coupling cosine terms; a constant forcing in $\dot{C}$; and the rational sine term in $\dot{C}$. The rational term $(PZ/S) \sin C$ is noteworthy because it involves division by a state variable, requiring careful library design to ensure representability.

3.1 Chaotic Dynamics and the Lyapunov Exponent

At the chosen parameters, the CLW system is chaotic. We estimate the maximal Lyapunov exponent via the QR decomposition method of Benettin et al. [1], integrating the variational equation for $T = 500$ time units after discarding a transient of $T_{\text{trans}} = 50$ time units, with QR renormalization every $\Delta t = 1.0$. This yields $\lambda_{\text{max}} \approx 0.3$, corresponding to a Lyapunov time of $T_\lambda = 1/\lambda_{\text{max}} \approx 3.3$ time units.

This implies that even a perfect model will exhibit $\mathcal{O}(1)$ trajectory deviations from the truth on timescales of a few Lyapunov times when initialized from nearby (but not identical) states. Consequently, we argue that trajectory-level metrics are informative only within $\sim 1\text{--}2T_\lambda$ and that equation-level metrics—support recovery and coefficient error—are the appropriate criteria for evaluating discovered models in this regime.

4 Methods

4.1 Data Generation

We generate training data using a “short burst” protocol: $n_{\text{traj}} = 250$ independent trajectories of length $T = 5.0$ time units, each integrated with a fourth/fifth-order Runge–Kutta method (SciPy `solve_ivp` with `rtol`= 10^{-9} , `atol`= 10^{-12}) at a time step of $\Delta t = 0.01$. Initial conditions are drawn uniformly from $[0.5, 2.0]^4$ for each state variable, providing broad coverage of the attractor basin.

The short-burst protocol avoids the need for a single long trajectory (which would sample the attractor statistically but provide less independent information per unit of data) and is more representative of experimental settings where repeated short acquisitions are available.

4.2 Noise Model

We corrupt the state measurements with additive Gaussian noise:

$$\tilde{\mathbf{X}} = \mathbf{X} + \eta \boldsymbol{\sigma}_X \odot \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad (5)$$

where $\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ is the noise amplitude, $\boldsymbol{\sigma}_X$ is the empirical standard deviation of the clean data computed per state variable, and $\odot$ denotes elementwise multiplication. This normalization ensures that η represents the noise-to-signal ratio uniformly across all four state variables.

4.3 Candidate Libraries

We consider three candidate libraries of increasing complexity:

Physics-informed library ($p = 10$ features). This library includes the constant function 1, the four state variables P , S , Z , the trigonometric functions $\cos C$ and $\sin C$, the three bilinear-trigonometric terms $ZS \cos C$, $ZP \cos C$, $PS \cos C$, and the rational-trigonometric term $(PZ/S) \sin C$. The ten functions span the CLW dynamics exactly.

Extended library ($p = 46$ features). Starting from the same ten basis functions, we augment the library with all pairwise products of the nine non-constant basis functions, yielding $10 + \binom{9}{2} + 9 = 10 + 36 = 46$ features. This tests recovery when the true dynamics are representable but embedded in a much larger candidate space.

Incomplete library ($p = 9$ features). We remove the rational term $(PZ/S) \sin C$ from the physics-informed library, leaving nine features. Because the true $\dot{C}$ equation contains this term, the dynamics are no longer representable. This ablation quantifies the penalty of library misspecification.

4.4 The SINDy Algorithm

We use the Sequential Thresholded Least Squares (STLSQ) optimizer [2] as implemented in PySINDy [7, 11]. STLSQ alternates between least-squares fitting and hard thresholding of small coefficients below a threshold λ . We set the ridge regularization parameter $\alpha = 0$ and do not normalize library columns.

Model selection. For each noise level η and seed, we sweep the STLSQ threshold over 25 logarithmically spaced values from 10^{-6} to 10^0 . The best model is selected by minimizing a penalized score:

$$\text{score} = \log(\text{MSE}) + w_{\text{nnz}} \cdot \text{nnz}, \quad (6)$$

where MSE is the mean squared prediction error on the training data, nnz is the number of nonzero coefficients, and $w_{\text{nnz}} = 2 \times 10^{-3}$ is the sparsity penalty weight. The constant term “1” is constrained to appear only in the $\dot{C}$ equation, reflecting the known structure.

4.5 Derivative Estimation Regimes

We compare three derivative estimation strategies of decreasing idealization:

1. **Oracle derivatives:** The exact right-hand side of the CLW system is evaluated at the (possibly noisy) state measurements. This is the most favorable scenario and serves as an upper bound on recovery quality.
2. **Numerical finite differences (FD):** Derivatives are computed from the noisy state measurements using second-order central differences via `numpy.gradient`. No smoothing is applied.
3. **Savitzky–Golay smoothed finite differences (SFD):** PySINDy’s `SmoothedFiniteDifference` method is used, which applies a Savitzky–Golay filter [16] to smooth the state data before differentiating. Unless otherwise noted, the default configuration uses a window length of 21, polynomial order 3, and differentiation order 2. In this regime, SINDy handles differentiation internally on a per-trajectory basis.

4.6 Evaluation Metrics

We evaluate discovered models using the following metrics:

Number of nonzero coefficients (nnz). The total count of nonzero entries in $\hat{\Xi}$. The true model has $k^* = 8$.

Coefficient relative L_2 error.

$$\epsilon_{\text{coef}} = \frac{\|\hat{\Xi} - \Xi^*\|_F}{\|\Xi^*\|_F}, \quad (7)$$

where Ξ^* is the ground-truth coefficient matrix aligned to the candidate library, and $\|\cdot\|_F$ denotes the Frobenius norm.

True positive rate (TPR). The fraction of truly nonzero coefficients that are correctly identified as nonzero: $\text{TPR} = |\hat{S} \cap S^*|/|\hat{S}|$, where S^* and $\hat{S}$ are the true and estimated support sets.

False positive rate (FPR). The fraction of truly zero coefficients that are incorrectly identified as nonzero: $\text{FPR} = |\hat{S} \setminus S^*|/|\hat{S}|$, where S^0 is the set of truly zero entries.

Exact support match. A binary indicator: $\hat{S} = S^*$.

Vector field error. The relative L_2 error of the model’s predicted right-hand side evaluated on the test data:

$$\epsilon_{\text{vf}} = \frac{\|\hat{f}(\mathbf{X}_{\text{test}}) - f^*(\mathbf{X}_{\text{test}})\|_F}{\|f^*(\mathbf{X}_{\text{test}})\|_F}. \quad (8)$$

All metrics are reported as means $\pm$ standard deviations over five independent random seeds.

5 Experimental Design

We organize our experiments into four groups:

Experiment 1: Derivative regime comparison. For each of the three derivative estimation methods (oracle, FD, SFD) and each noise level $\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$, we fit a SINDy model and evaluate all metrics. This experiment isolates the effect of derivative quality on recovery.

Experiment 2: Library ablation. We compare the physics-informed, extended, and incomplete libraries. In the noiseless case with oracle derivatives, both the physics-informed and extended libraries achieve exact recovery (Table 1). Under noise, the extended library is tested with oracle derivatives across the full η sweep to assess how library size interacts with noise (Table 2). The incomplete library is tested similarly to quantify misspecification effects.

Table 1: Clean recovery with oracle derivatives (mean $\pm$ std, 5 seeds). Both libraries achieve exact symbolic recovery.

Library	nnz	ϵ_{coef}	TPR	FPR	Exact	ϵ_{vf}
Physics-informed ($p=10$)	8 ± 0	3.76×10^{-16}	1.0	0.0	5/5	4.08×10^{-16}
Extended ($p=46$)	8 ± 0	3.76×10^{-16}	1.0	0.0	5/5	4.08×10^{-16}

Experiment 3: Sample-size ablation. At a fixed moderate noise level $\eta = 0.01$ with oracle derivatives and the extended library, we sweep the number of training trajectories $n_{\text{traj}} \in \{25, 50, 100, 250, 500\}$ to assess data efficiency.

Experiment 4: SFD sensitivity. We sweep the SFD window length $w \in \{5, 11, 21, 41\}$ and polynomial order $q \in \{2, 3, 5\}$ at three noise levels $\eta \in \{10^{-3}, 10^{-2}, 10^{-1}\}$ to characterize sensitivity to smoothing hyperparameters.

6 Results

6.1 Exact Recovery in the Noiseless Regime

With clean data and oracle derivatives, both the physics-informed (10 features) and extended (46 features) libraries achieve exact symbolic recovery (Table 1). The discovered coefficient matrix matches the ground truth to machine precision ($\epsilon_{\text{coef}} < 4 \times 10^{-16}$), with $\text{nnz} = 8 = k^*$, $\text{TPR} = 1.0$, $\text{FPR} = 0.0$, and exact support match in all five seeds. This confirms that STLSQ can identify the correct sparse structure from the candidate library when given noiseless data and perfect derivatives, even when the library is substantially overspecified.

6.2 Derivative Regime Comparison Under Noise

Table 2 presents the central result: a comparison of the three derivative estimation regimes across noise levels. Figures 1 and 2 visualize the coefficient error and TPR trends.

Several patterns emerge:

Low noise ($\eta \leq 10^{-2}$). All three regimes recover all true terms ($\text{TPR} \geq 0.93$) with varying accuracy. The SFD regime achieves the lowest coefficient error ($\epsilon_{\text{coef}} = 0.026\text{--}0.112$), demonstrating the benefit of smoothing when noise is moderate relative to the signal. The numerical FD regime achieves excellent results at $\eta = 10^{-3}$ ($\epsilon_{\text{coef}} = 0.027$), comparable to SFD. The oracle regime shows higher variance at low noise due to its sensitivity to the noise in the state through which it evaluates the RHS.

Moderate noise ($\eta = 10^{-1}$). The SFD method degrades sharply, retaining only $\text{TPR} = 0.33$ with $\text{nnz} = 2.6$, indicating excessive sparsification. The oracle and FD regimes retain more structure ($\text{TPR} = 0.88$ and 0.58 , respectively), though coefficient accuracy is substantially degraded.

Table 2: Consolidated derivative regime comparison (mean $\pm$ std, 5 seeds). Physics-informed library ($p=10$).

Regime	η	nnz	ϵ_{coef}	TPR	FPR	ϵ_{vf}
Oracle	10^{-4}	10.6	0.187	1.00	0.08	0.363
	10^{-3}	10.2	0.469	1.00	0.07	0.843
	10^{-2}	12.0	0.395	0.93	0.14	0.362
	10^{-1}	12.2	0.470	0.88	0.16	0.448
	1	4.0	1.007	0.13	0.09	0.962
	10	0.8	0.935	0.10	0.00	0.990
FD	10^{-4}	12.0	0.117	1.00	0.13	0.203
	10^{-3}	10.4	0.027	1.00	0.08	0.030
	10^{-2}	11.2	0.346	0.98	0.11	0.350
	10^{-1}	6.2	0.625	0.58	0.05	0.555
	1	2.8	1.079	0.08	0.07	0.988
	10	6.8	3.665	0.03	0.21	1.842
SFD	10^{-4}	11.0	0.026	1.00	0.09	0.024
	10^{-3}	11.0	0.027	1.00	0.09	0.025
	10^{-2}	11.4	0.112	1.00	0.11	0.084
	10^{-1}	2.6	0.711	0.33	0.00	0.685
	1	3.8	0.942	0.35	0.03	0.779
	10	2.4	1.436	0.08	0.06	1.058

High noise ($\eta \geq 1$). Recovery fails across all regimes. The oracle and SFD regimes converge to $\text{nnz} \approx 1\text{--}4$ with $\epsilon_{\text{coef}} \approx 1.0$, indicating that the discovered model is no better than the trivial zero predictor (for which $\epsilon_{\text{coef}} = 1$ by definition). The FD regime at $\eta = 10$ exhibits $\epsilon_{\text{coef}} = 3.67$ and $\epsilon_{\text{vf}} = 1.84$, reflecting catastrophic error amplification by finite differences on heavily corrupted data.

6.3 Library Ablation

Extended library under noise. Table 3 compares the physics-informed (oracle) and extended (oracle) libraries under noise. In the clean case, both achieve exact recovery. Under noise, the extended library shows comparable coefficient error but elevated FPR (0.06–0.19 vs. 0.07–0.16), reflecting spurious term inclusion from the larger candidate space. At high noise ($\eta \geq 1$), both libraries fail similarly.

Incomplete library. The incomplete library, missing $(PZ/S) \sin C$, floors the coefficient error at $\epsilon_{\text{coef}} \approx 0.48$ regardless of noise level (Table 3). This floor reflects the irreducible error from the missing term: the $\dot{C}$ equation cannot be represented, and the optimizer compensates by distributing the unrepresented dynamics across available features, yielding persistently elevated FPR ($\sim 0.17\text{--}0.19$) even at $\eta = 10^{-4}$. The TPR remains 1.0 for $\eta \leq 0.1$ (all recoverable terms are found), confirming that the error is localized to the missing term.

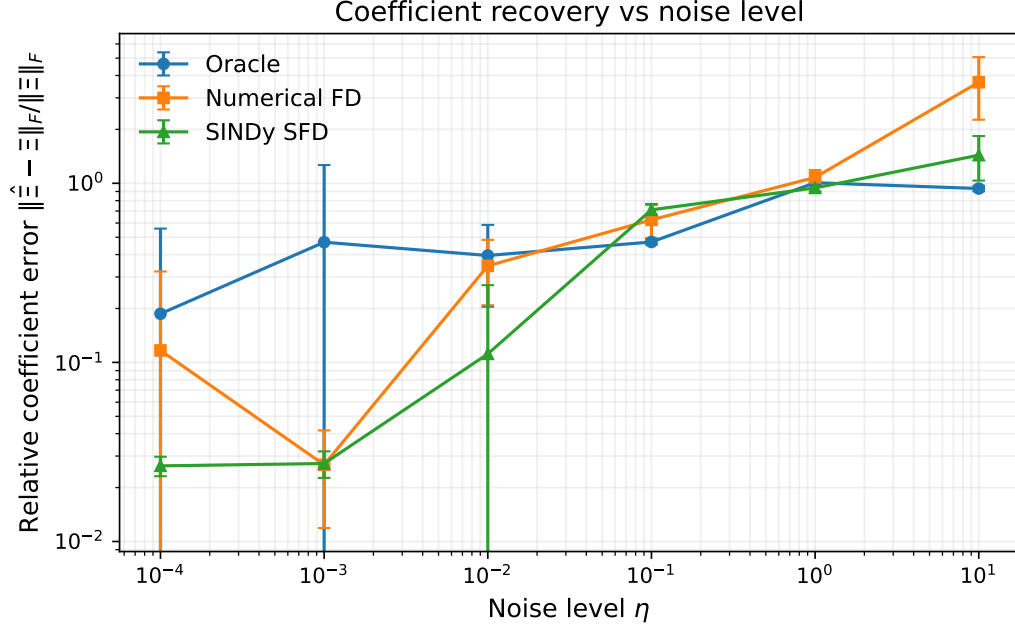


Figure 1: Coefficient relative L_2 error versus noise level η for the three derivative estimation regimes. SFD provides the best low-noise recovery but degrades sharply at $\eta \geq 0.1$.

6.4 Sample-Size Ablation

Table 4 reports the effect of training set size at fixed $\eta = 0.01$ with oracle derivatives and the extended library. All configurations maintain $\text{TPR} \geq 0.975$, but the coefficient error and FPR vary substantially. At $n_{\text{traj}} = 25$, the best results are obtained ($\epsilon_{\text{coef}} = 0.107$, exact support in 1/5 seeds), while larger datasets show higher coefficient errors due to increased false-positive inclusion (FPR increases from 0.10 to 0.15). This counterintuitive result likely reflects the interaction between threshold selection and sample size: with more data, weaker spurious correlations survive the STLSQ threshold.

6.5 SFD Sensitivity

Figure 4 shows the SFD hyperparameter sensitivity. The key findings are:

- At low noise ($\eta = 10^{-3}$), short windows ($w = 5$) perform best, achieving the lowest ϵ_{coef} (~ 0.011) and exact support match in 2/5 seeds. Excessively long windows ($w = 41$) introduce smoothing bias, inflating nnz to ~ 20 and ϵ_{coef} to ~ 0.11 – 0.14 .
- At moderate noise ($\eta = 10^{-2}$), intermediate windows ($w = 11$ – 21) provide the best balance, with $\epsilon_{\text{coef}} \approx 0.02$ – 0.05 and $\text{TPR} = 1.0$.
- At high noise ($\eta = 10^{-1}$), no SFD configuration recovers more than $\text{TPR} = 0.88$, and coefficient error exceeds 0.48 across all window/order combinations. The Savitzky–Golay filter cannot rescue recovery at this noise level.

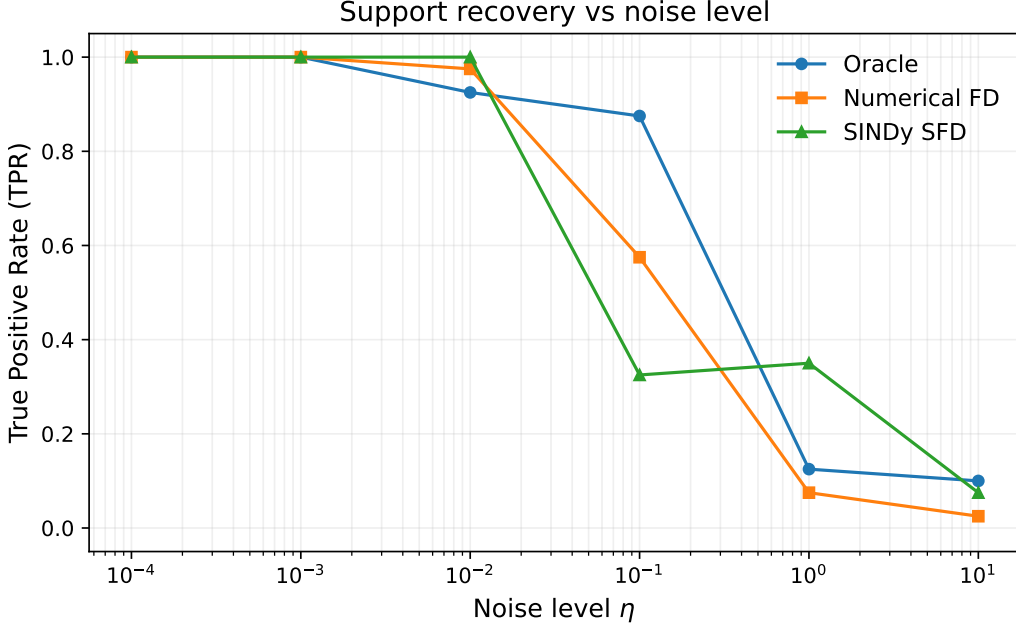


Figure 2: True positive rate versus noise level η . All regimes maintain $\text{TPR} = 1.0$ for $\eta \leq 0.01$; SFD degrades most rapidly at higher noise.

- Higher polynomial orders ($q = 5$) are generally beneficial at larger windows, mitigating the smoothing bias that afflicts low-order fits.

6.6 The Chaos Evaluation Problem

Figure 5 illustrates the fundamental evaluation challenge for chaotic systems: two trajectories of the CLW system initialized with $\Delta C = 10^{-6}$ diverge to $\mathcal{O}(1)$ differences within ~ 10 time units ($\sim 3T_\lambda$). This divergence occurs for the *true model*, not a discovered approximation, confirming that trajectory-level error at horizons beyond $\sim 1-2T_\lambda$ reflects the sensitivity to initial conditions rather than model quality.

Figure 6 shows trajectory error normalized by Lyapunov time for models discovered in the three derivative regimes. In the SFD regime at low noise ($\eta \leq 10^{-2}$), the discovered model tracks the truth for $\sim 1-2T_\lambda$ before exponential divergence, consistent with a near-exact model. At higher noise, the initial error is larger and divergence sets in within a fraction of T_λ .

7 Discussion

7.1 Summary of Findings

Our experiments establish a clear hierarchy of difficulty for sparse equation discovery on the CLW system:

Table 3: Library ablation: physics-informed vs. extended vs. incomplete (oracle derivatives, mean over 5 seeds).

Library	η	nnz	ϵ_{coef}	TPR	FPR	ϵ_{vf}
Physics ($p=10$)	10^{-4}	10.6	0.187	1.00	0.08	0.363
	10^{-2}	12.0	0.395	0.93	0.14	0.362
	10^{-1}	12.2	0.470	0.88	0.16	0.448
Extended ($p=46$)	10^{-4}	10.0	0.495	1.00	0.06	0.960
	10^{-2}	11.6	0.291	0.98	0.12	0.305
	10^{-1}	12.0	0.468	0.88	0.16	0.447
Incomplete ($p=9$)	10^{-4}	12.4	0.480	1.00	0.19	0.444
	10^{-2}	11.6	0.493	0.94	0.17	0.444
	10^{-1}	12.0	0.421	1.00	0.17	0.447

Table 4: Sample-size ablation ($\eta = 0.01$, oracle derivatives, extended library, 5 seeds).

n_{traj}	nnz	ϵ_{coef}	TPR	FPR	Exact
25	11.2 ± 2.3	0.107 ± 0.162	1.00	0.10	1/5
50	12.2 ± 2.6	0.235 ± 0.202	1.00	0.13	1/5
100	11.8 ± 2.6	0.231 ± 0.204	1.00	0.12	1/5
250	11.8 ± 1.3	0.330 ± 0.169	0.98	0.13	0/5
500	12.6 ± 2.1	0.376 ± 0.150	0.98	0.15	0/5

1. **Noiseless, oracle derivatives:** Exact recovery is achievable with both physics-informed and overly rich libraries. STLSQ reliably selects the correct 8-term support from up to 46 candidate features.
2. **Low noise** ($\eta \leq 10^{-2}$): All derivative regimes maintain $\text{TPR} \geq 0.93$. SFD provides the best coefficient accuracy, reducing ϵ_{coef} by an order of magnitude relative to oracle-on-noisy-state at $\eta = 10^{-3}$.
3. **Moderate noise** ($\eta = 10^{-1}$): The SFD regime undergoes a phase transition, collapsing to ~ 2.6 terms. Oracle and FD regimes retain more structure but with degraded accuracy.
4. **High noise** ($\eta \geq 1$): Recovery fails uniformly. The coefficient error approaches or exceeds 1.0 (no better than zero), and support identification is effectively random.

7.2 The Role of Library Design

The library ablation reveals two complementary failure modes:

- **Overspecification** (extended library): Exact recovery is preserved in the noiseless case, confirming the theoretical identifiability. Under noise, the additional degrees of

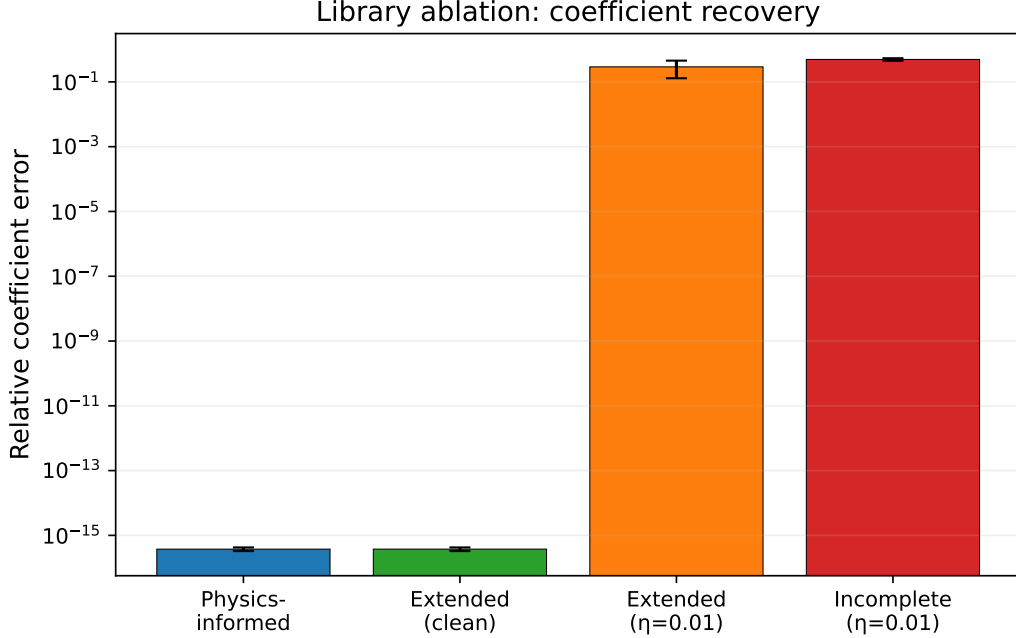


Figure 3: Library ablation: coefficient error across noise levels for the three library types. The incomplete library has a persistent error floor at ~ 0.48 due to the missing rational term.

freedom provide more opportunities for spurious term inclusion, modestly increasing FPR.

- **Misspecification** (incomplete library): The missing rational term creates an irreducible error floor of $\epsilon_{\text{coef}} \approx 0.48$ that dominates at all noise levels below the general failure regime ($\eta \geq 1$). Importantly, this floor manifests even at $\eta = 10^{-4}$, where the physics-informed library achieves near-exact recovery.

These results underscore that library design is at least as critical as noise handling for practical equation discovery.

7.3 SFD: Benefits and Limitations

The Savitzky–Golay smoothed finite difference method provides a substantial improvement over raw finite differences at low-to-moderate noise, achieving the best coefficient accuracy among the three regimes for $\eta \leq 10^{-2}$. However, it exhibits a sharp “cliff” at $\eta \approx 10^{-1}$, where the smoothing becomes insufficient to suppress the noise-induced derivative error, and the STLSQ threshold eliminates most true terms.

The SFD sensitivity analysis reveals that the window length is the dominant hyperparameter. Short windows ($w = 5$) minimize bias at low noise but provide insufficient smoothing at higher noise. Long windows ($w = 41$) introduce smoothing bias that inflates nnz regardless of noise level. Intermediate windows ($w = 11$ – 21) provide the most robust performance across noise levels.

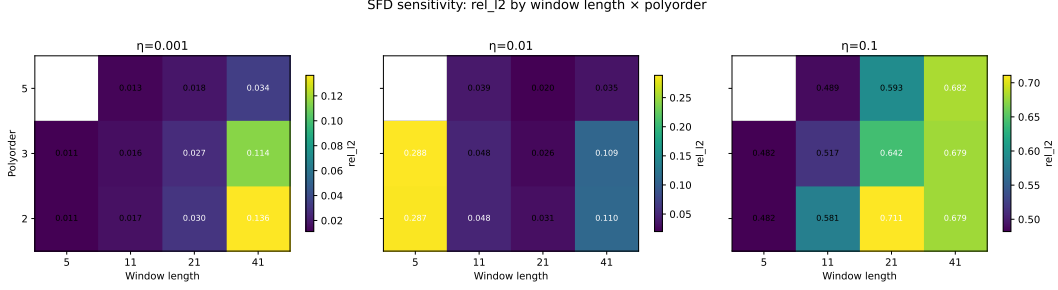


Figure 4: SFD sensitivity: coefficient error as a function of window length and polynomial order, at three noise levels.

7.4 Limitations

We acknowledge several limitations of this study:

- **Single system:** Our experiments are restricted to the CLW model at one parameter setting. While this provides a complete controlled study, the results may not generalize to all chaotic ODE systems or to systems with qualitatively different nonlinearities.
- **Reduced model:** The CLW system is a low-dimensional reduction of tokamak turbulence dynamics. Real plasma dynamics involve spatial degrees of freedom, multiple time scales, and sources of noise beyond additive Gaussian state corruption.
- **Known library structure:** The physics-informed library was designed with knowledge of the true dynamics. In practice, a domain expert may not know the relevant nonlinearities *a priori*; the extended-library results provide some guidance for this scenario.
- **Additive Gaussian noise:** Our noise model is idealized. Experimental plasma diagnostics produce noise with potentially non-Gaussian, state-dependent, and temporally correlated structure.
- **No online or adaptive methods:** We use a fixed pipeline (data $\rightarrow$ derivatives $\rightarrow$ STLSQ). Iterative or Bayesian approaches that adapt the library or noise model during fitting [8] may extend the recoverable noise range.

8 Conclusion

We have presented a systematic study of sparse equation discovery applied to the Chen–Lin–White reduced tokamak turbulence model. Our key findings are:

1. **Exact recovery is achievable:** Given clean oracle derivatives and a library containing the true terms (even if substantially overspecified), STLSQ recovers the exact CLW equations with coefficient error below 10^{-15} .

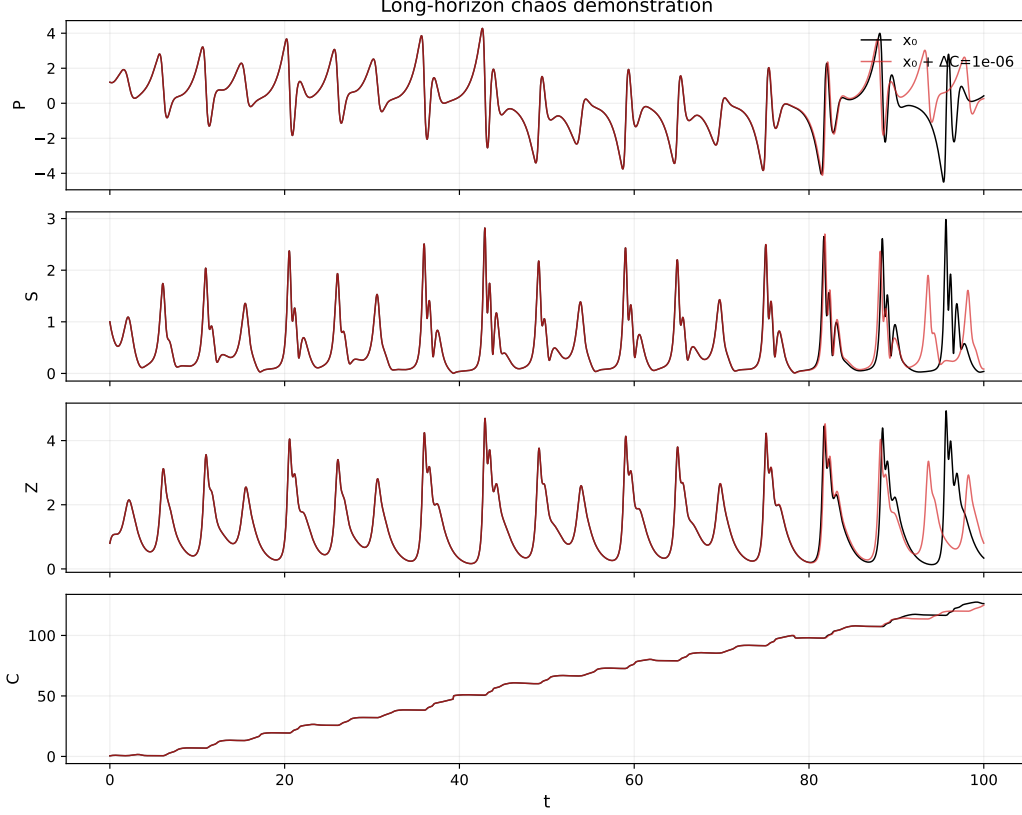


Figure 5: Chaos demonstration: two CLW trajectories with $\Delta C = 10^{-6}$ in initial conditions. The *true model* produces $\mathcal{O}(1)$ deviations within $\sim 3T_\lambda$, exposing the inadequacy of long-horizon trajectory error as a model quality metric.

2. **Noise tolerance has clear boundaries:** All derivative estimation regimes maintain useful recovery ($\text{TPR} \geq 0.93$, $\epsilon_{\text{coef}} < 0.4$) for $\eta \leq 10^{-2}$, with SFD providing the best accuracy in this range. At $\eta \geq 10^{-1}$, recovery degrades substantially, and at $\eta \geq 1$, it fails entirely.
3. **Library design is critical:** An incomplete library floors the coefficient error at ~ 0.48 regardless of noise level, while an overly rich library modestly increases false-positive risk. The practical implication is that ensuring the library contains the correct terms is more important than minimizing its size.
4. **Chaos requires equation-level evaluation:** For chaotic systems like CLW, trajectory comparison beyond ~ 1 – 2 Lyapunov times ($T_\lambda \approx 3.3$) reflects sensitivity to initial conditions rather than model quality. We advocate for support recovery and coefficient error as the primary evaluation criteria.

The CLW model, despite its low dimensionality, captures the essential challenges of sparse equation discovery in chaotic, fusion-relevant systems: nonlinear coupling, rational nonlinearities, and the chaos evaluation problem. Our results provide a quantitative baseline

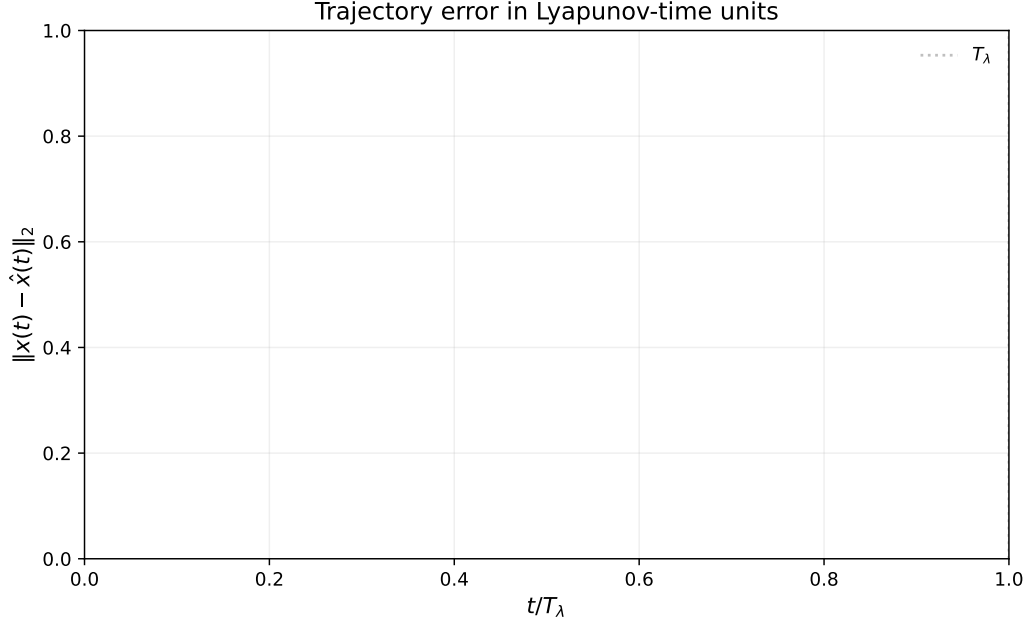


Figure 6: Trajectory error vs. time (in Lyapunov units $T_\lambda \approx 3.3$) for discovered models in the three derivative regimes at selected noise levels. Low-noise SFD models track the truth for $\sim 1\text{--}2T_\lambda$ before the inevitable chaotic divergence.

for future work on more complex plasma turbulence models, higher-dimensional systems, and more sophisticated sparsity-promoting algorithms.

Code Availability

All code and data to reproduce the experiments in this paper are available at [https://github.com/\[repository\]](https://github.com/[repository]).

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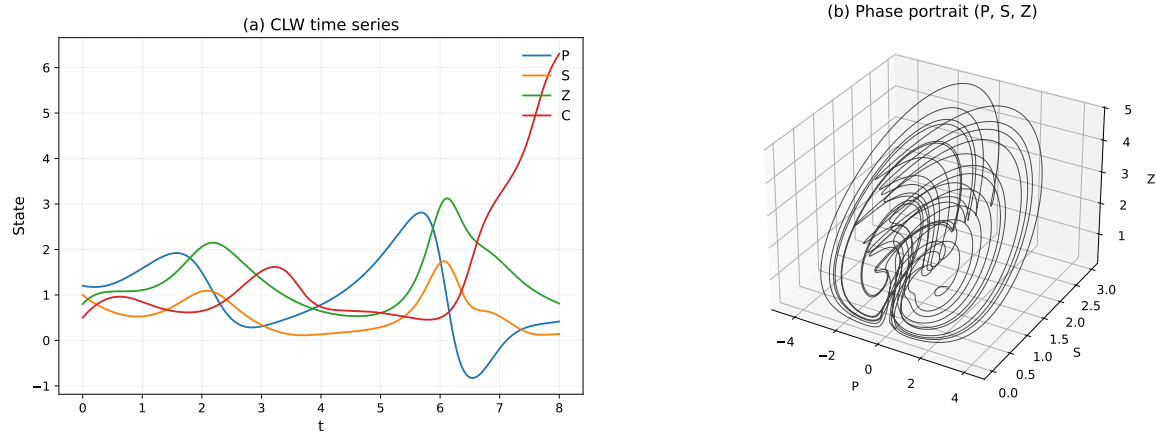


Figure 7: CLW system overview. (Left) Time series of the four state variables P , S , Z , C . (Right) Three-dimensional phase portrait in (P, S, Z) space, colored by time, showing the chaotic attractor.

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